25900015 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION, MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBTS100 - PREPARATION OF BIOLOGICAL SPECIMENS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. What is a transect in ecological sampling? [U] / [CO1]

   a). A soil testing instrument

   b). A straight line or path along which observations are made

   c). A map used for navigation

   d). A type of laboratory experiment

2. Why is glycerin sometimes used in fish specimen preservation? [U] / [CO1, CO2]

   a). To prevent drying

   b). To add weight

   c). To speed up decay

   d). To change texture

3. Which of the following is an example of a simple stain? [K] / [CO4]

   a). Hematoxylin

   b). Eosin

   c). Methylene Blue

   d). Safranin

4. Why is it called an "orange peel" bucket? [K] / [CO1]

   a). Its design resembles the opening and closing of orange peels

   b). Because it is used for harvesting oranges

c). Due to its bright orange color

d). It is used in citrus processing plants

5. Which type of mounting medium is commonly used for preparing permanent slides? [U] / [CO4]

a). Water

b). Glycerin

c). Canada balsam

d). Oil immersion

6. Which of the following is a reason for using an articulated skeleton in museums or classrooms? [K] / [CO3]

a). It serves as a permanent anatomical reference.

b). It helps study the biomechanics and posture of an animal.

c). It provides a three-dimensional view of bone arrangement.

d). All of the above

7. The respiratory organs in arthropods can be [U] / [CO1, CO2]

a). Gills

b). Tracheae

c). Book lungs

d). All of the above

8. Why is tissue embedding essential before using a rotary microtome? [U] / [CO3]

a). To prevent the tissue from drying

b). To improve staining quality

c). To provide structural support and stability for sectioning

d). None of the above

9. Which of the following organisms belongs to the phylum Coelenterata? [U] / [CO1, CO2]

a). Planaria

b). Earthworm

c). Hydra

d). Ascaris

10. Which complex tissue is responsible for the transport of water and minerals in plants? [U] / [CO4]

a). Phloem

b). Xylem

c). Periderm

d). Epidermis

11. Which of the following is an example of a parasitic nematode? [U] / [CO1, CO2]

a). Ascaris

b). Tapeworm

c). Planaria

d). Amoeba

12. What is the primary excretory organ in Mollusca? [U] / [CO1, CO2]

a). Nephridia b). Malpighian tubules

c). Kidneys d). Flame cells

[1x12 = 12]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **3** marks

13. What are the main stages in the life cycle of a frog? [K] / [CO3]

14. Explain the importance of fixation in museum specimen preparation. [U] / [CO5]

15. Name two methods of specimen preservation. [K] / [CO1]

16. Who is responsible for enforcing specimen collection laws? [K] / [CO1]

17. What are the two main groups of angiosperms based on their seed structure, and give an example of each? [K] / [CO4]

18. What is the role of airtight containers in wet-preserved plant storage? [U] / [CO5]

19. What adaptations help reptiles survive in terrestrial environments? [U] / [CO1, CO2]

20. How does the season affect plant collection? [U] / [CO5]

[3x6 = 18]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

21. What are biological specimen preservatives, and why are they necessary for research and study? [U] / [CO1]

22. Explain how taxidermy is used in the preservation of mammal specimens. [U] / [CO1, CO2]

23. What precautions should be taken while storing herbarium sheets to prevent damage? [K] / [CO5]

24. What challenges do researchers face while collecting plant specimens in remote areas? [U] / [CO5]

25. Briefly classify Phylum Porifera and provide suitable examples for each group. [K] / [CO1, CO2]

26. What is the process of embedding biological specimens in resin, and what materials are commonly used? [K] / [CO3]

[5x4 = 20]